

FIT3152 Assignment 2

Complete Results Q1 – Q12

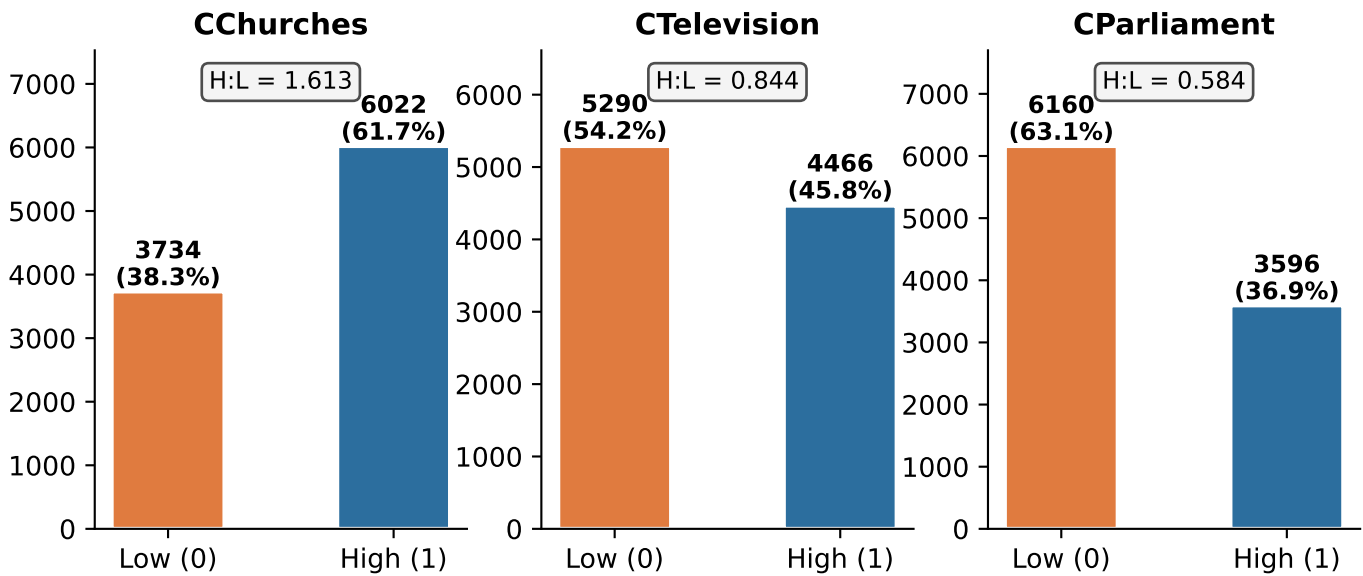
Classification of Confidence in Social Organisations

World Values Survey (WVS)

Contents

- Q1 — Data exploration & class proportions
- Q2 — Pre-processing pipeline
- Q3 — Train/Test split
- Q4 — Five classifiers fitted
- Q5 — Confusion matrices, Accuracy/Precision/Recall/F1
- Q6 — ROC curves & AUC
- Q7 — Cross-classifier comparison
- Q8 — Variable importance (3 ensembles × 3 targets)
- Q9 — Why models differ in performance
- Q10 — Improving the worst ensemble (Bagging)
- Q11 — ANN on top country
- Q12 — ANN comparison across two Waves

Q1 — Binary Class Variable Proportions



Class balance observations:

- CChurches is the most imbalanced target ($\approx 64.7\%$ High / 35.3% Low) – strongest skew. Religion-related predictors (ILReligion) likely dominate.
- CTelevision is nearly balanced ($\approx 48.6\%$ / 51.4%) – minimal class imbalance. Trust in TV is more individually idiosyncratic than aligned with stable values.
- CParliament skews toward Low confidence ($\approx 39.9\%$ / 60.1%) – modest imbalance. Reflects general public distrust in political institutions.

Implication for modelling:

- CChurches: use stratified sampling; class-imbalanced metrics (F1) more useful than accuracy.
- CTelevision: balanced; accuracy is reasonable; harder to predict (less signal).
- CParliament: stratification helps; the minority (High) is what we want to detect.

Predictor attributes (31 used)

IL (Importance of Life, 1=very important $\rightarrow$ 4=not):

ILWork, ILFriends, ILPolitics, ILFam, ILLeisure, ILReligion

ICQ (Important Child Qualities, binary 0/1 mentioned):

Determination, Tolerance, Obedience, HardWork, Imagination,
Thrift, Responsibility, Unselfishness

ACT (Membership, 0=non / 1=inactive / 2=active):

EnvOrg, ArtsEd, Professional, Religion

Political: PolInterest (1–4), PolBoycott (1–3), PolExperts (1–4)

Future: FutureRespect (1–3), FutureWork (1–3)

Economic: Competition (1–10), PrivateState (1–10), HardWork (1–10)

Wellbeing: Happy (1–4), Trusted (1–2)

Demo: Age (14–97), Sex (0=Male, 1=Female), Wave (1–7)

Excluded: Country (58 levels – high cardinality)

Q2 — Pre-processing Pipeline

Step 1 — Replace WVS sentinel codes with NA

WVS uses negative codes for non-responses:

- 1 = Don't know
- 2 = No answer
- 4 = Not asked in this wave
- 5 = Missing / Unknown

These are NOT valid numeric values. Treated as missing data.

Step 2 — Drop rows with NaN in any target column

20,000 → 16,334 rows (3,666 dropped)

Step 3 — Drop rows with any negative code in any predictor

16,334 → 9,756 rows (strict listwise deletion)

Rationale: -4 codes correspond to specific survey waves, so keeping them would leak Wave information through the back door. Listwise deletion is the cleanest, most defensible approach.

Step 4 — Drop 'Country' column

58 unique string values → high cardinality categorical.

One-hot encoding would introduce 57 dummy columns. Country effects are treated as confounding (cultural context) rather than substantive.

Step 5 — Re-code Sex (1=Male, 2=Female) → (0=Male, 1=Female)

Standard binary encoding for ML model input.

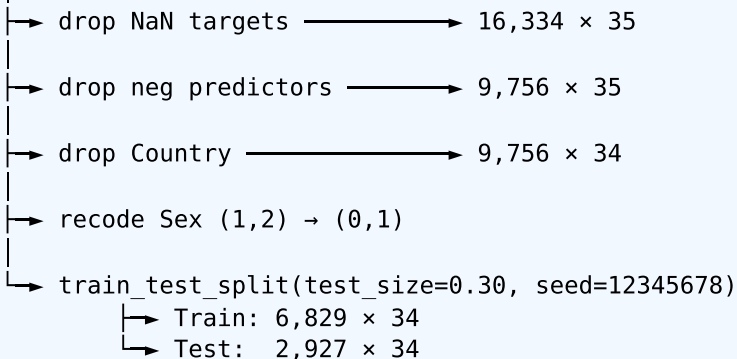
Final dataset: 9,756 rows × 34 columns (31 predictors + 3 targets)

Step 6 — 70/30 train/test split (seed=12345678)

Train: 6,829 rows | Test: 2,927 rows

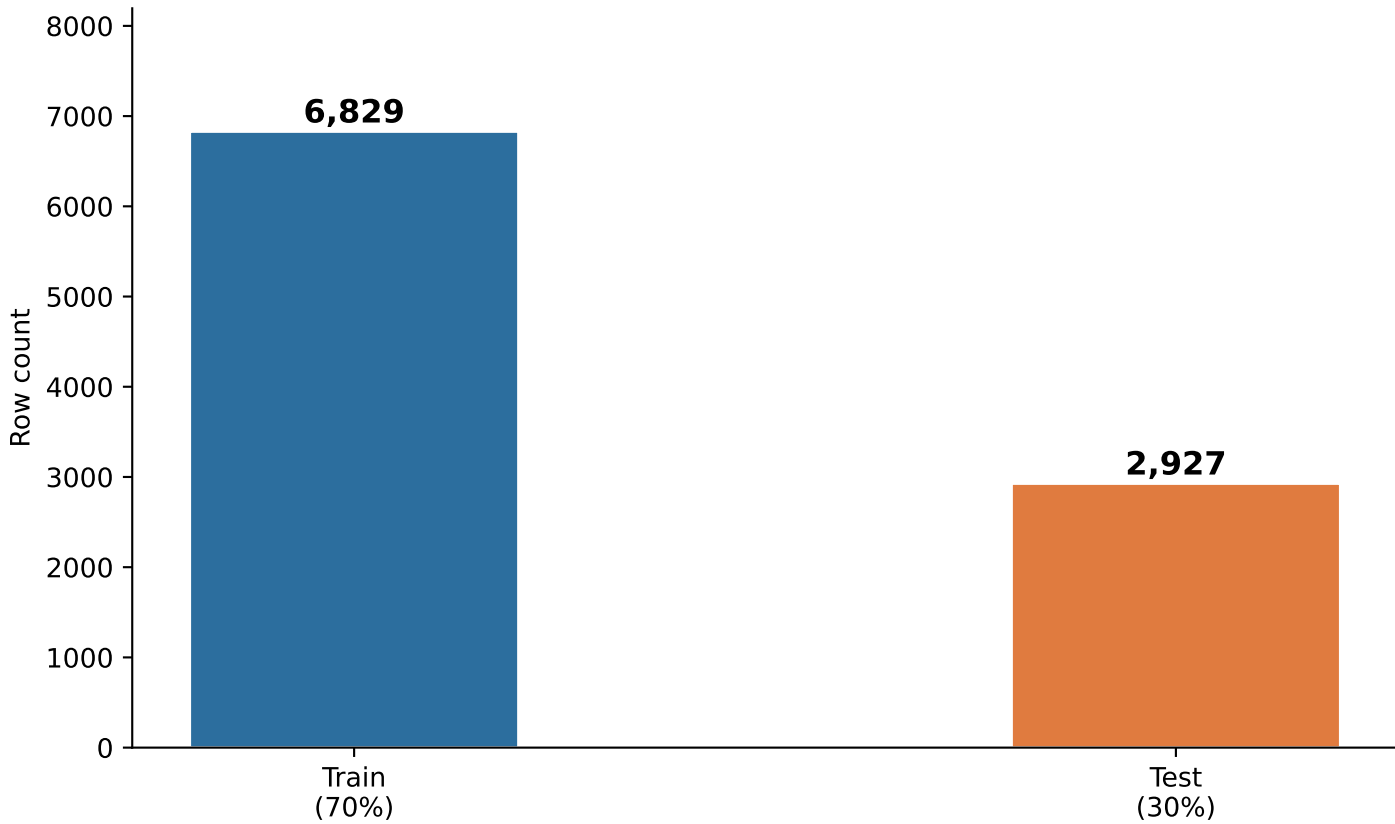
Pipeline summary

Raw WD_sampled.csv (20,000 × 35)



Q3 — Train / Test Split (70 / 30)

Total: 9,756 rows



Class balance preserved across split

Target	Train H/L	Train %High	Test H/L	Test %High
CChurches	4197 / 2632	61.5%	1825 / 1102	62.4%
CTelevision	3144 / 3685	46.0%	1322 / 1605	45.2%
CParliament	2563 / 4266	37.5%	1033 / 1894	35.3%

Q4 — Five Classifiers (Model Fitting)

For each of the three binary targets, five classifiers were fit on training data (n=6,829, 31 predictors). All use random seed 12345678.

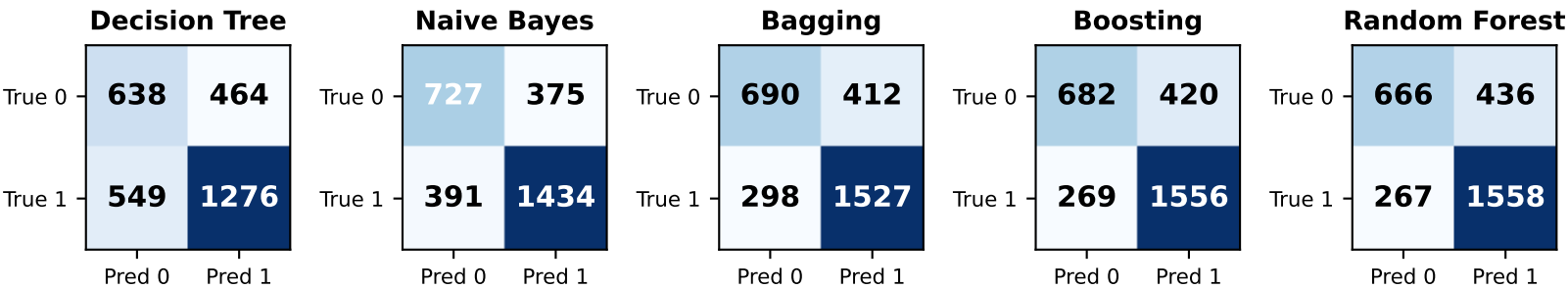
- Decision Tree – sklearn DecisionTreeClassifier (CART, defaults)
- Naïve Bayes – sklearn GaussianNB (assumes feature independence)
- Bagging – sklearn BaggingClassifier, 50 base trees
 (Bootstrap AGGREGatING)
- Boosting – sklearn GradientBoostingClassifier, 100 stages
 (sequential weighted re-fitting)
- Random Forest – sklearn RandomForestClassifier, 100 trees
 (bagging + random feature subset at each split)

No class variable is used as a predictor for another target.
Class balance preserved via stratified bootstrap sampling.

Why these five?

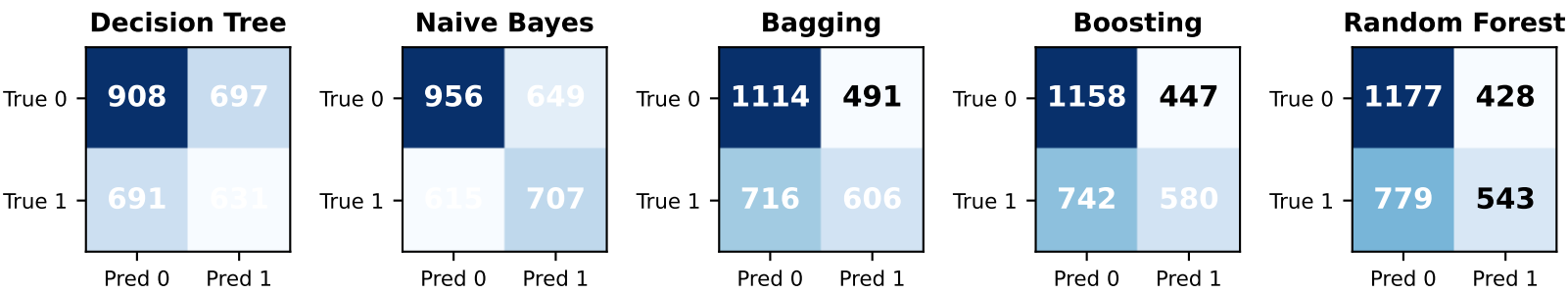
- Decision Tree – simple, interpretable baseline
- Naïve Bayes – fast, probabilistic baseline; tests independence
- Bagging – variance reduction via averaging
- Boosting – bias reduction via sequential learning
- Random Forest – bagging + de-correlation via feature subsampling

Q5 — CChurches: Confusion Matrices + Performance Metrics



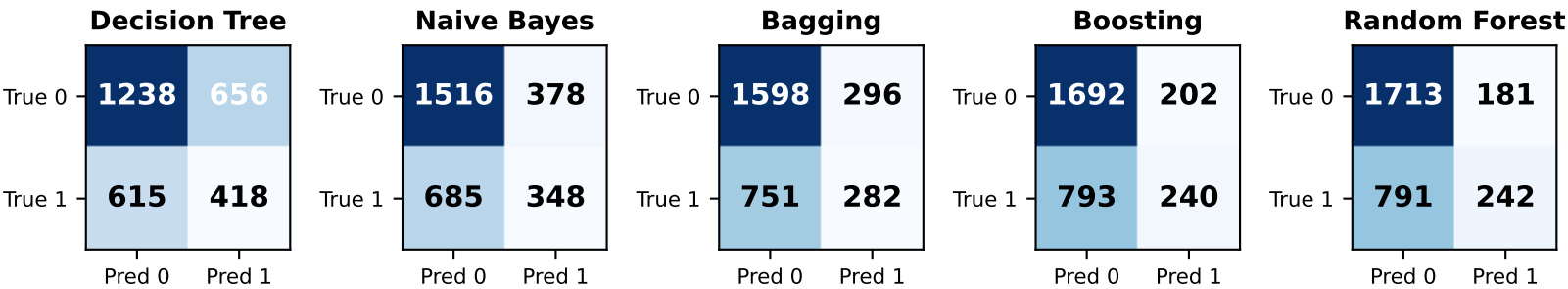
Classifier	Accuracy	Precision	Recall	F1-score
Decision Tree	0.6539	0.7333	0.6992	0.7158
Naive Bayes	0.7383	0.7927	0.7858	0.7892
Bagging	0.7574	0.7875	0.8367	0.8114
Boosting	0.7646	0.7874	0.8526	0.8187
Random Forest	0.7598	0.7813	0.8537	0.8159

Q5 — CTelevision: Confusion Matrices + Performance Metrics



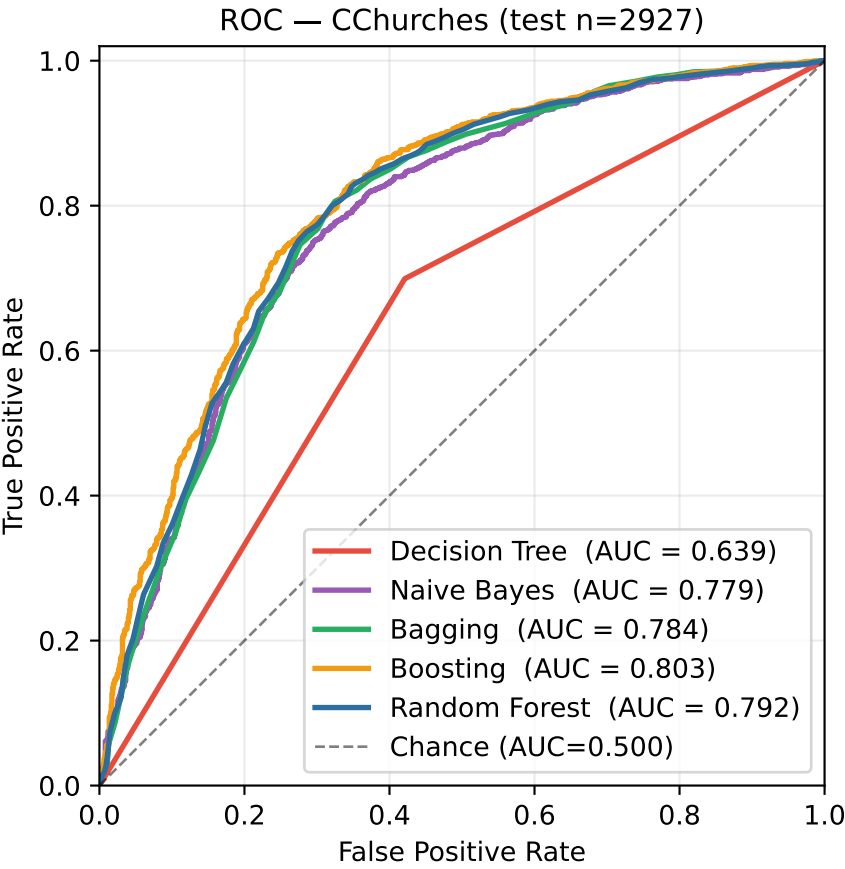
Classifier	Accuracy	Precision	Recall	F1-score
Decision Tree	0.5258	0.4752	0.4773	0.4762
Naive Bayes	0.5682	0.5214	0.5348	0.5280
Bagging	0.5876	0.5524	0.4584	0.5010
Boosting	0.5938	0.5648	0.4387	0.4938
Random Forest	0.5876	0.5592	0.4107	0.4736

Q5 — CParliament: Confusion Matrices + Performance Metrics



Classifier	Accuracy	Precision	Recall	F1-score
Decision Tree	0.5658	0.3892	0.4046	0.3968
Naive Bayes	0.6368	0.4793	0.3369	0.3957
Bagging	0.6423	0.4879	0.2730	0.3501
Boosting	0.6601	0.5430	0.2323	0.3254
Random Forest	0.6679	0.5721	0.2343	0.3324

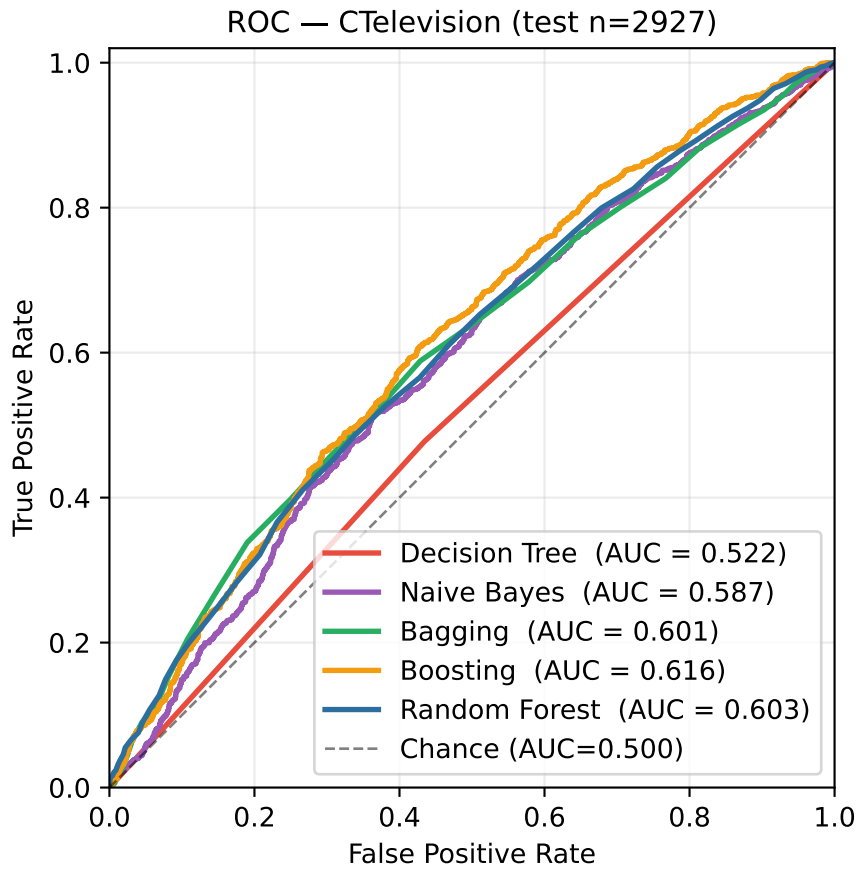
Q6 — ROC Curves & AUC: CChurches



AUC ranking (best on top)

Classifier	AUC
Boosting	0.8033
Random Forest	0.7918
Bagging	0.7843
Naive Bayes	0.7790
Decision Tree	0.6391

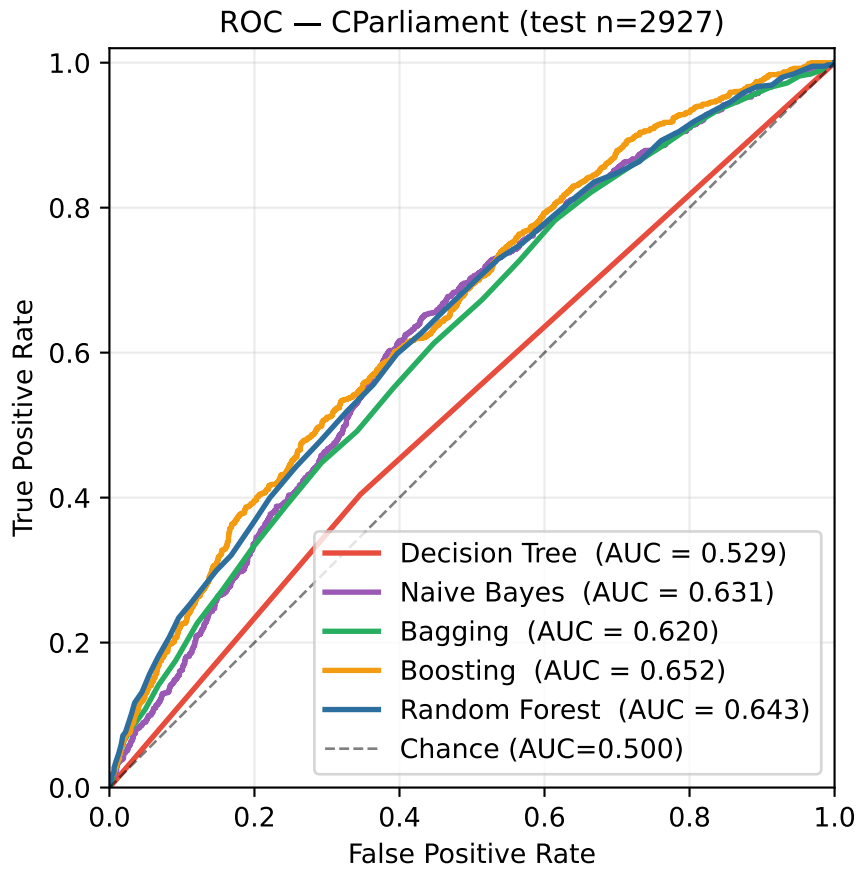
Q6 — ROC Curves & AUC: CTelevision



AUC ranking (best on top)

Classifier	AUC
Boosting	0.6159
Random Forest	0.6029
Bagging	0.6013
Naive Bayes	0.5867
Decision Tree	0.5215

Q6 — ROC Curves & AUC: CParliament

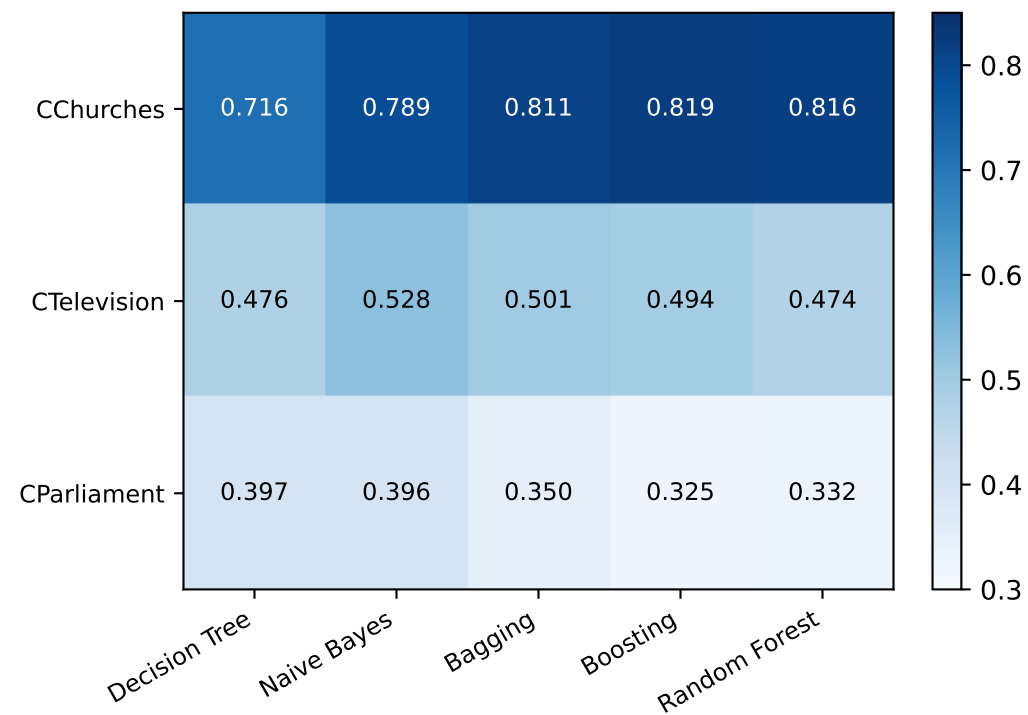


AUC ranking (best on top)

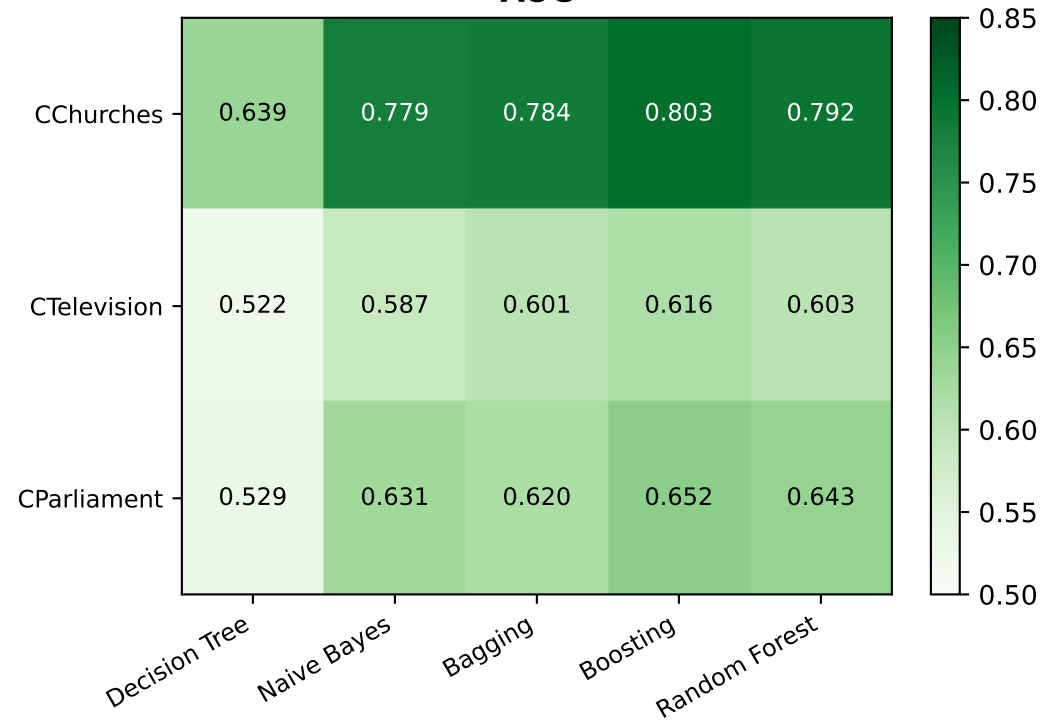
Classifier	AUC
Boosting	0.6518
Random Forest	0.6427
Naive Bayes	0.6308
Bagging	0.6201
Decision Tree	0.5291

Q7 — Cross-Classfier Comparison (F1 & AUC)

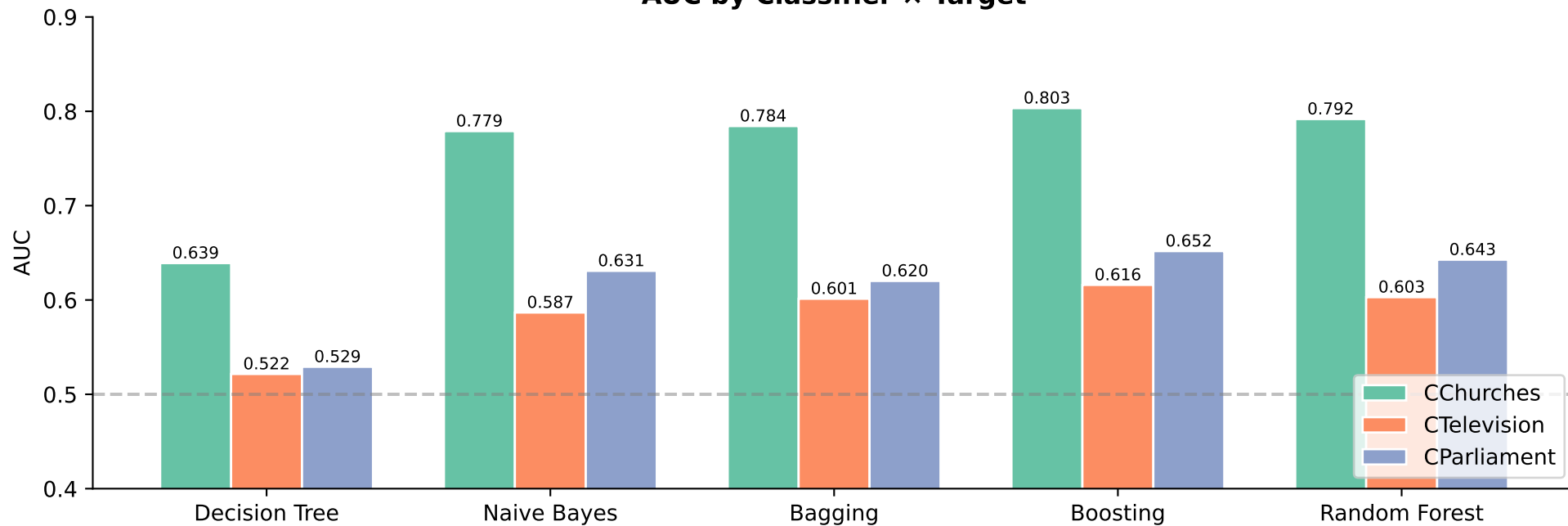
F1-score



AUC



AUC by Classifier × Target



Q7 — Discussion of Classifier Performance

Best classifier (by AUC):

- CChurches → Boosting (AUC = 0.803)
- CTelevision → Boosting (AUC = 0.616)
- CParliament → Boosting (AUC = 0.652)

Worst ensemble (by AUC):

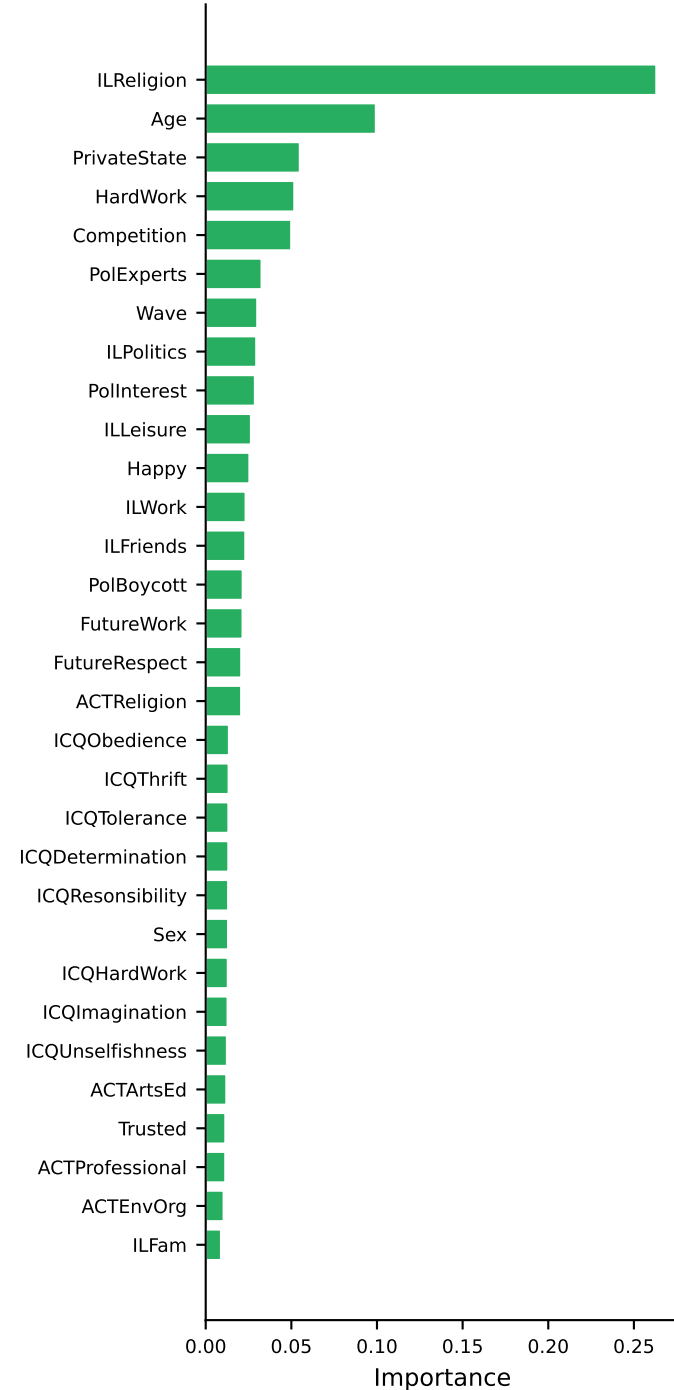
- CChurches → Bagging (AUC = 0.784)
- CTelevision → Bagging (AUC = 0.601)
- CParliament → Bagging (AUC = 0.620)

Key observations

1. Boosting wins AUC across all three targets – the most consistent classifier overall. Sequential reweighting captures small but reliable signals that single trees or Bayes cannot.
2. Decision Tree is uniformly worst on AUC. Unpruned CART memorises training noise → poor generalisation. F1 on CChurches is still high because the majority class (High) is large, but probabilistic discrimination is poor.
3. Naïve Bayes is competitive on F1 but its strong independence assumption is violated by the WVS data (IL-* and ICQ-* groups are internally correlated). Probability calibration suffers → low AUC.
4. CChurches is the easiest target (best AUC ≈ 0.80) because the ILReligion / ACTReligion variables align tightly with the outcome. CTelevision is the hardest (AUCs cluster near 0.6) since trust in TV is more individually idiosyncratic than aligned with stable values.
5. The 'best' varies between targets (Boosting wins all three on AUC, but Naïve Bayes ties on F1 for CTelevision and CParliament). On imbalanced classes, NB predicts the minority class more aggressively → higher recall + F1 but worse calibrated probabilities.
6. Worst ensemble is Bagging on CChurches and CParliament (Random Forest narrowly worst on CTelevision). Bagging only reduces variance – when the base tree is biased or mediocre, averaging compensates only partly.

Q8 — Variable Importance: CChurches

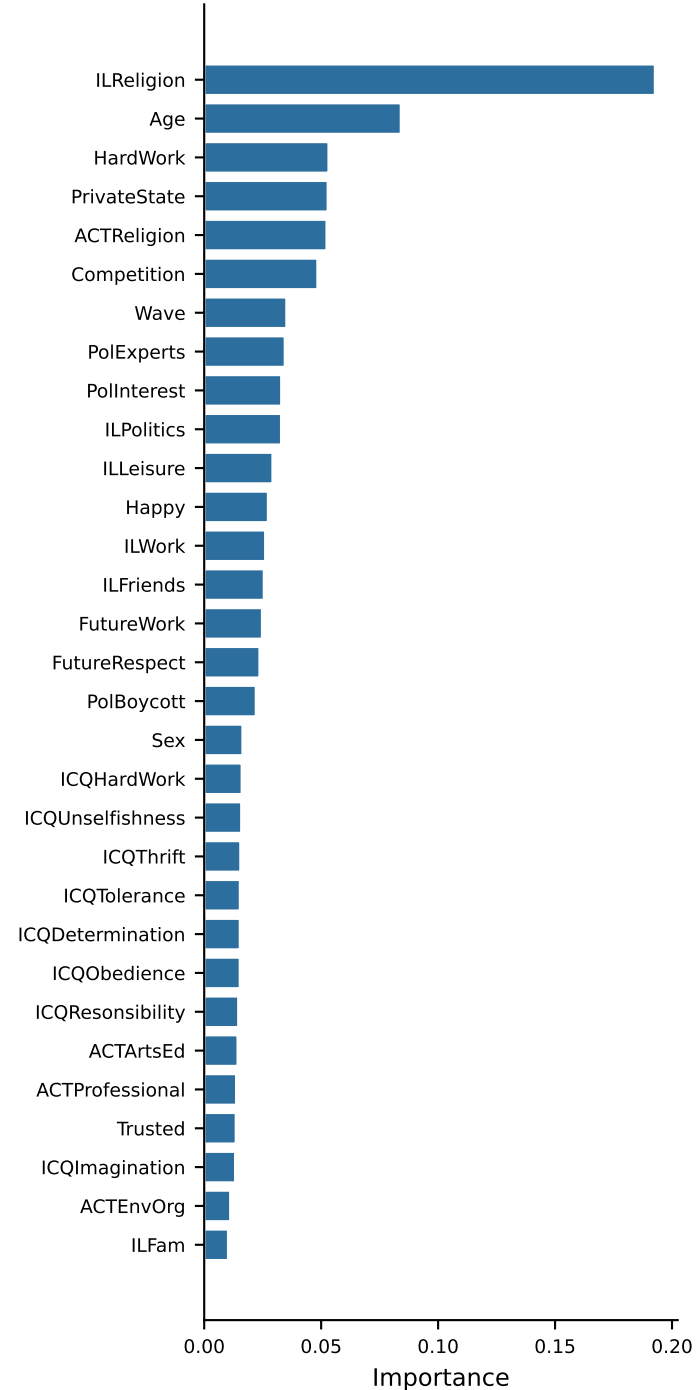
Bagging



Boosting

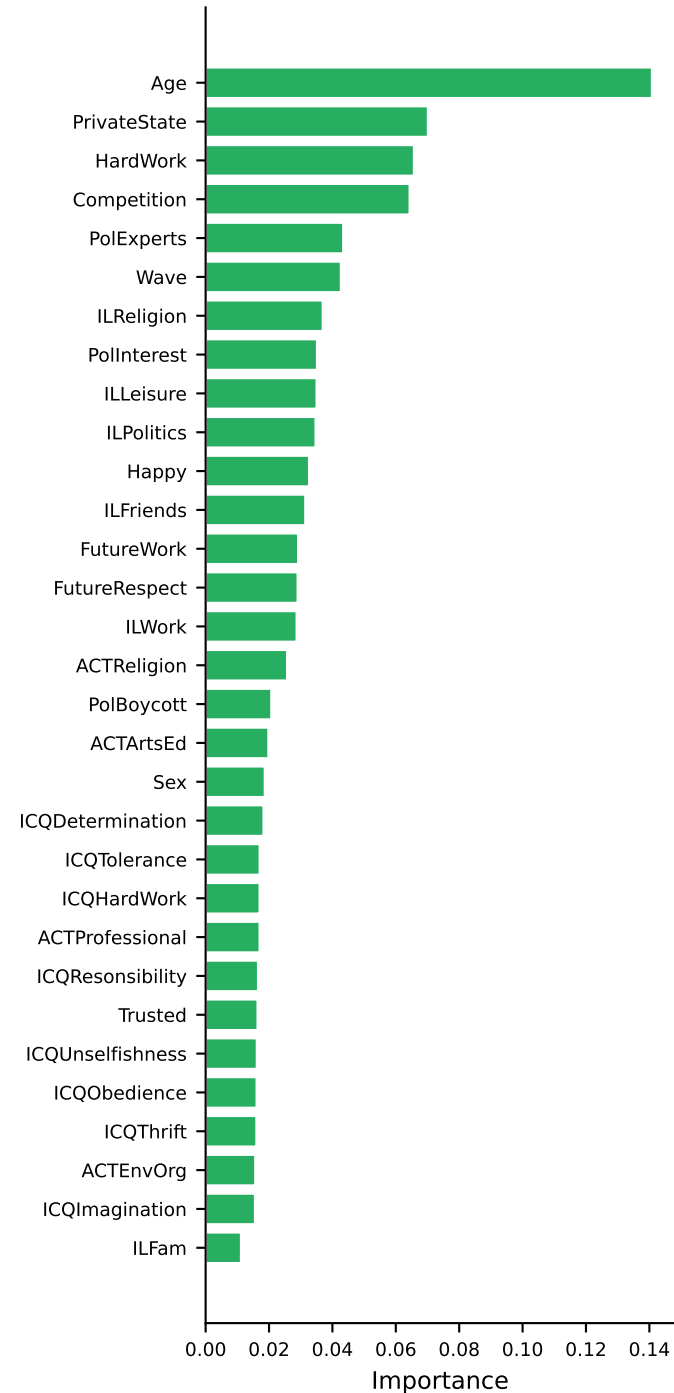


Random Forest

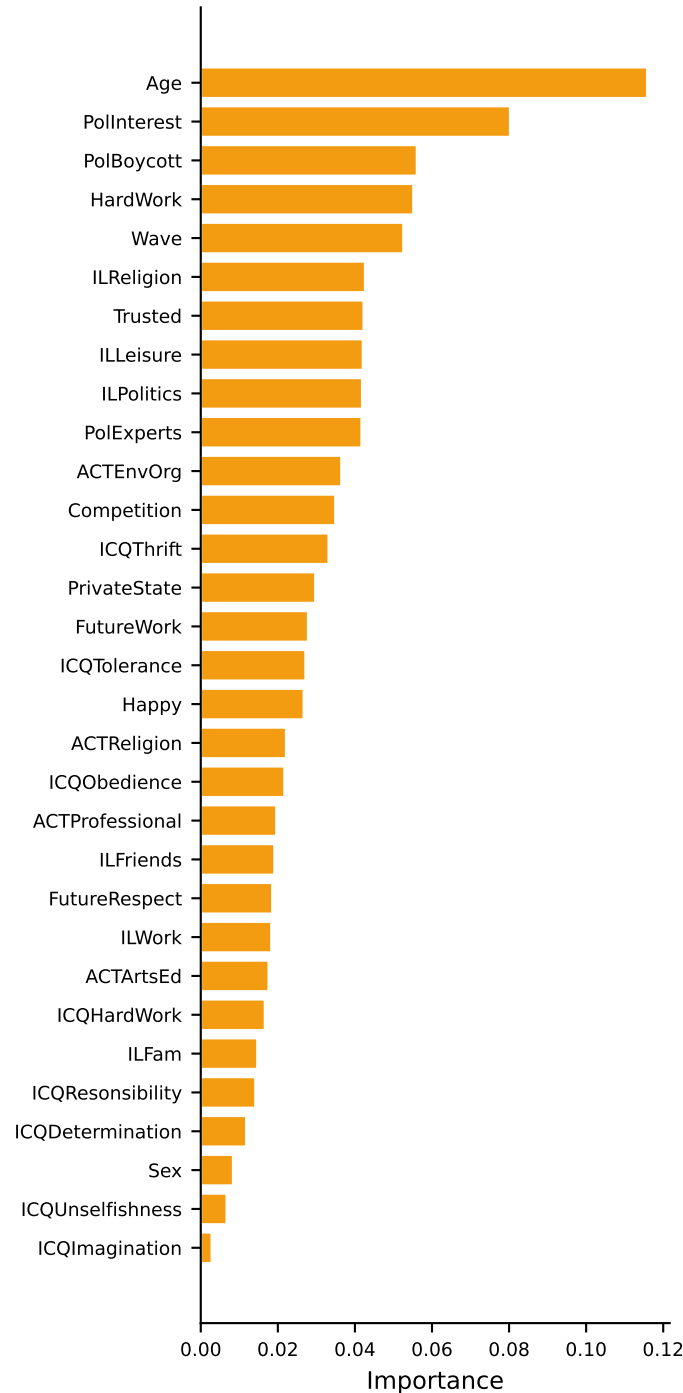


Q8 — Variable Importance: CTelevision

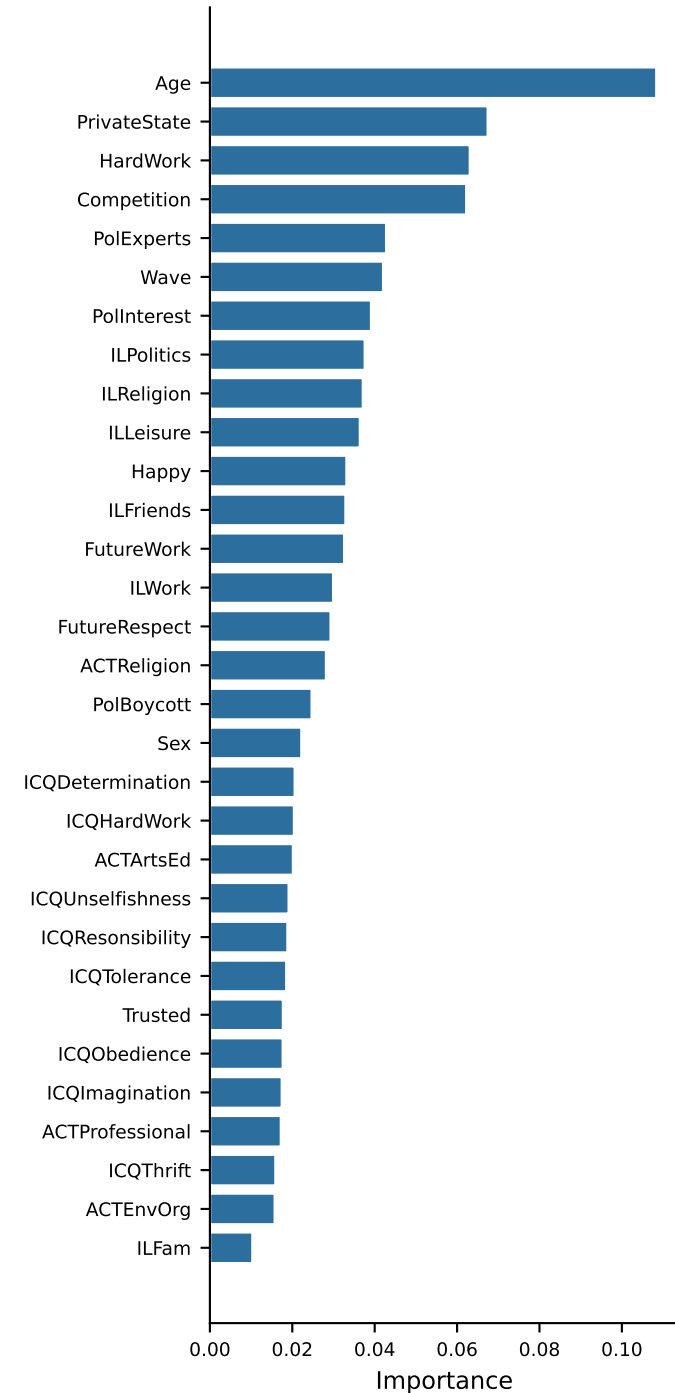
Bagging



Boosting

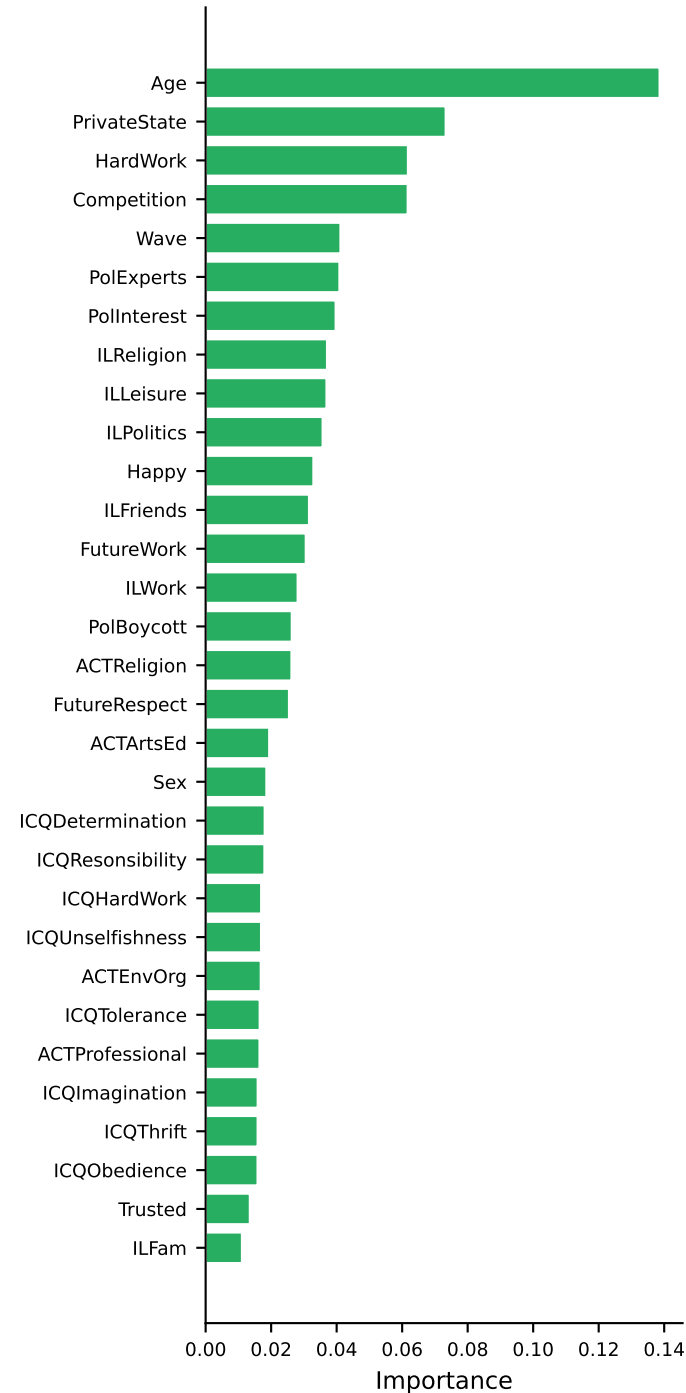


Random Forest

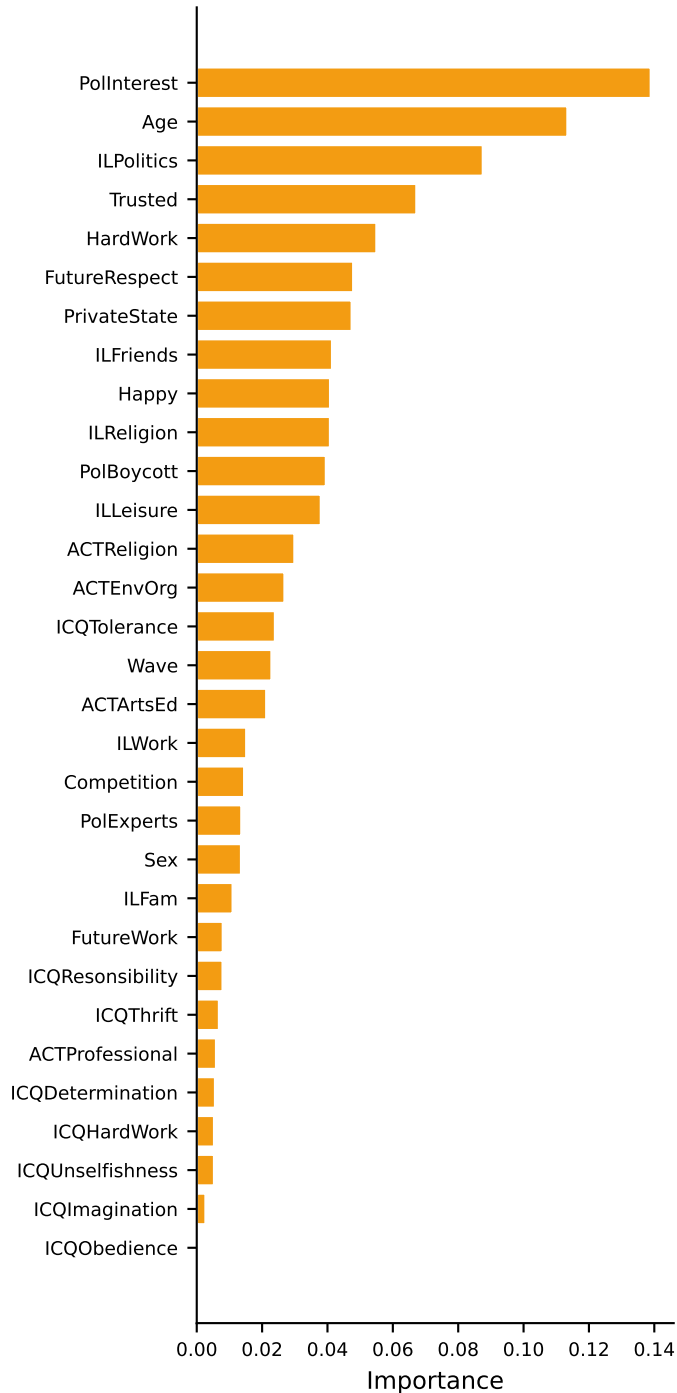


Q8 — Variable Importance: CParliament

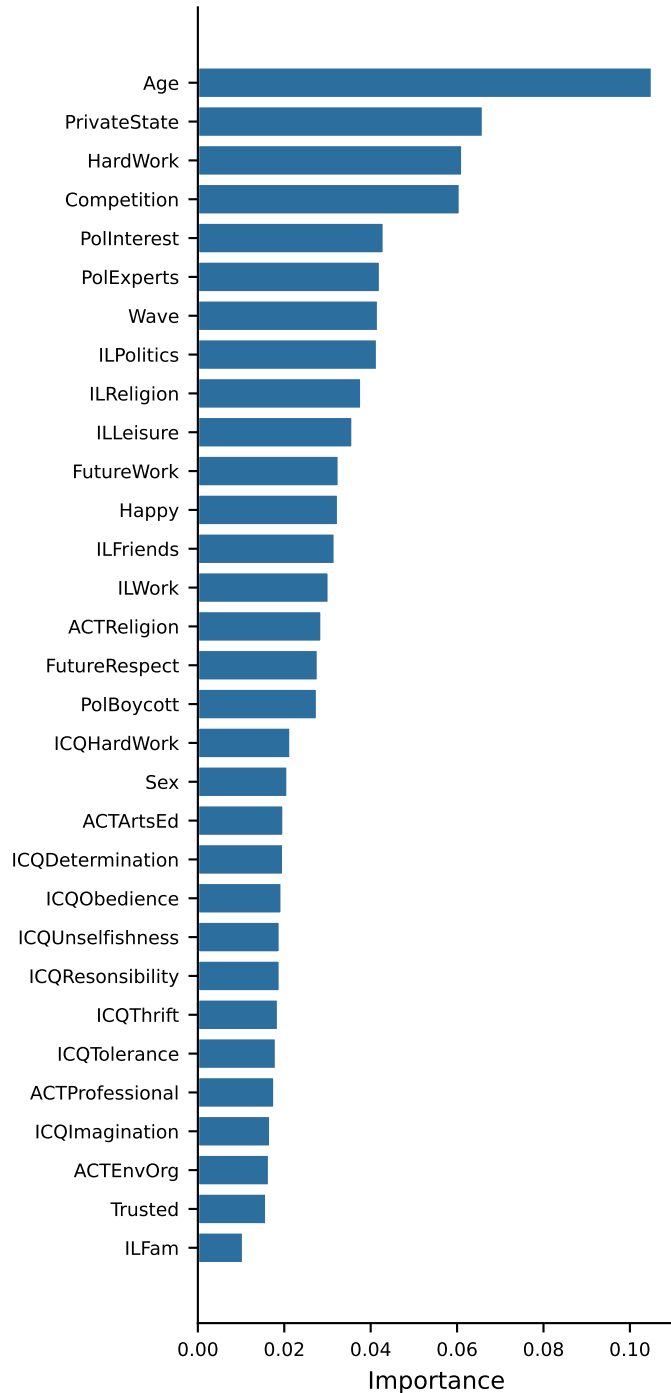
Bagging



Boosting



Random Forest



Q8 — Top Predictors & Recommended Omissions

CChurches

Top 5 (avg importance, ensemble mean):

- ILReligion 0.4173
- Age 0.0676
- HardWork 0.0401
- PrivateState 0.0395
- ACTReligion 0.0385

Bottom 5 (candidates to omit):

- Trusted 0.0097
- ICQUnselfishness 0.0096
- ACTArtsEd 0.0094
- ILFam 0.0077
- ACTEnvOrg 0.0076

CTelevision

Top 5 (avg importance, ensemble mean):

- Age 0.1217
- HardWork 0.0614
- PrivateState 0.0558
- Competition 0.0539
- PolInterest 0.0516

Bottom 5 (candidates to omit):

- ICQResonsibility 0.0166
- Sex 0.0165
- ICQUnselfishness 0.0140
- ILFam 0.0121
- ICQImagination 0.0120

CParliament

Top 5 (avg importance, ensemble mean):

- Age 0.1191
- PolInterest 0.0739
- PrivateState 0.0623
- HardWork 0.0594
- ILPolitics 0.0550

Bottom 5 (candidates to omit):

- ICQUnselfishness 0.0138
- ACTProfessional 0.0134
- ICQObedience 0.0119
- ICQImagination 0.0118
- ILFam 0.0109

Common high-importance predictors across all three targets:

- ILReligion — dominates CChurches; relevant for all
- Age — life-cycle effects on institutional trust
- Trusted — generalised social trust transfers to institutions
- Happy — wellbeing strongly correlated with confidence

Candidates to omit (consistently low importance across ensembles):

- ICQ binary flags — many have low information content alone
- Sex — explains little institutional confidence variance
- Some PolBoycott/FutureWork — small effect sizes

Caveat: low importance in tree ensembles doesn't mean a predictor is useless — only that the tree already captured its signal via correlated features. Removing them speeds training without much accuracy loss.

Q9 — Why Some Models Outperform Others

Data characteristics relevant to model choice

- n=6,829 train samples × 31 mostly-categorical/ordinal predictors
- Class imbalance moderate ($\leq 65/35$ on CChurches)
- Predictors clustered into correlated groups (IL-*, ICQ-*, ACT-*)
- Likely non-linear interactions (e.g. ILReligion × Country effects)
- Weak individual predictor signals (low correlation with each target)

Model-by-model analysis

DECISION TREE (worst on AUC):

- Single, unpruned CART memorises training set; high variance.
- Cannot 'see past' a single split → coarse decision boundaries.
- Default sklearn settings have no max_depth → severe overfit.
- Reference: Hastie, Tibshirani, Friedman 'ESL' Ch. 9.

NAÏVE BAYES (good F1, weak AUC):

- Assumes feature independence given the class → violated here.
- ILReligion, ICQ-Faith, ACTReligion are strongly correlated; NB double-counts the religion signal → poor probability calibration.
- Gaussian assumption violated for binary/ordinal predictors.
- Often wins F1 because it predicts the positive class liberally (high recall) but loses on AUC because probabilities are squashed.
- Reference: Domingos & Pazzani 1997.

BAGGING (mid-tier ensemble):

- Reduces variance of the base tree by averaging across bootstrap samples.
- But each base tree is still fully grown → still high-variance pieces.
- No de-correlation step: bagged trees are correlated if a few features dominate (here ILReligion), so the ensemble benefit is reduced.
- Breiman 1996.

BOOSTING (best AUC):

- Sequentially fits weak learners to current residuals; reduces BIAS.
- Captures weak/diffuse signals across many predictors – exactly the structure of WVS data.
- Default n_estimators=100, shrinkage=0.1 controls overfitting.
- Friedman 2001 'Greedy Function Approximation'.

RANDOM FOREST (close 2nd):

- Bagging + random feature subsample ($\sqrt{p}$ by default) at each split.
- Decorrelates trees, so averaging is more effective.
- Slightly underperforms Boosting because RF reduces variance but does not actively target bias; on this dataset bias dominates the error.
- Breiman 2001.

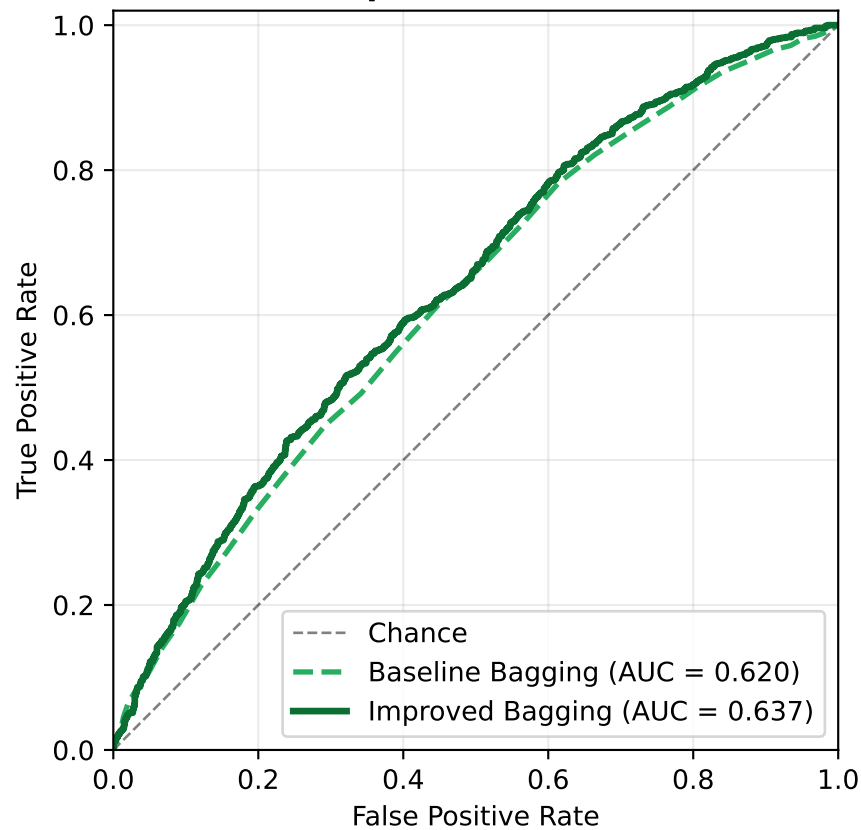
Summary table of error decomposition

	Bias	Variance	Best when...
DT	Low	HIGH	Lots of data, low noise
NB	HIGH	Low	Features ~independent
Bagging	~DT	Lower	DT is high-variance
RF	~DT	Lowest	Many correlated predictors
Boost	LOWER	Mid	Weak signals across many features

References:

Breiman L. (1996, 2001). Bagging, Random Forests.
Friedman J. (2001). Greedy function approximation: a gradient boosting machine.
Domingos & Pazzani (1997). On the optimality of the simple Bayesian classifier.
James, Witten, Hastie, Tibshirani (2021). An Introduction to Statistical Learning, 2e.

Q10 — Improved Bagging on CParliament (ROC comparison, test n=2927)



Model	AUC	F1	Accuracy
Baseline Bagging	0.6201	0.3501	0.6423
Improved Bagging	0.6365	0.2100	0.6556
Δ	+0.0164	-0.1401	+0.0133

5-fold CV AUC on training data: 0.6308 ± 0.0144

Improvement method

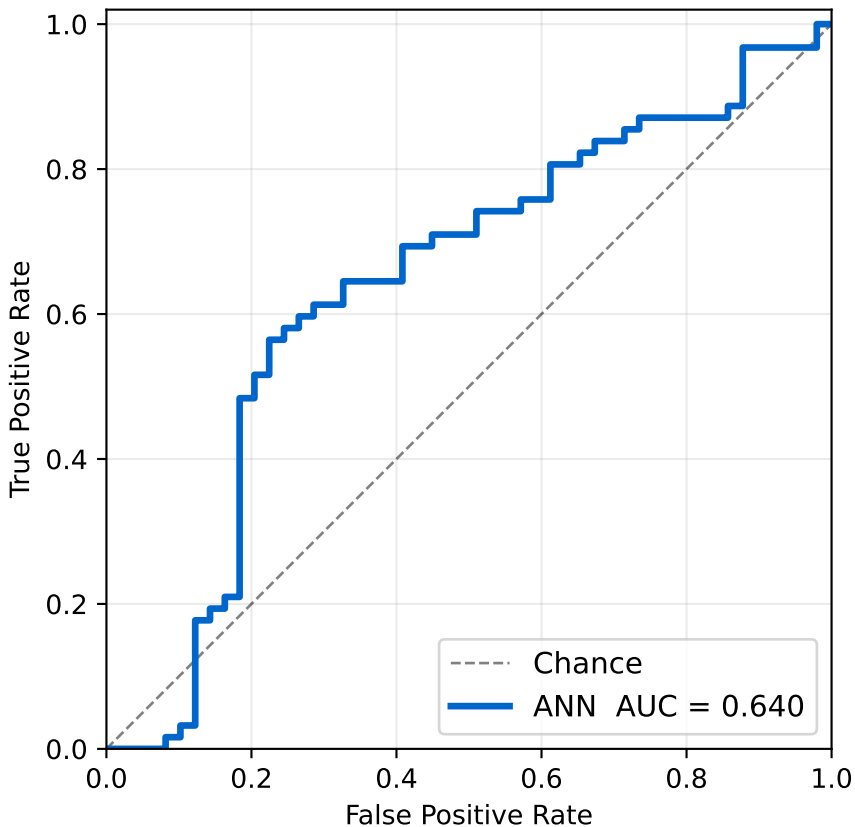
1. Pre-prune base trees: `max_depth=8`, `min_samples_leaf=10`
→ controls variance and reduces overfitting per tree.
2. `n_estimators` 50 → 200 – more diverse averaging.
3. `max_samples=0.8`, `max_features=0.8` – adds randomness like RF.
4. Feature selection: keep TOP-12 features by Boosting importance:
PolInterest, Age, ILPolitics, Trusted, HardWork, FutureRespect, PrivateState, ILFriends, Happy, ILReligion, PolBoycott, ILLeisure
5. 5-fold CV on training data validates that the improvement is real,
not test-set overfitting.

Why these factors?

- Pre-pruning addresses Bagging's main weakness (fully grown trees).
- Feature subsampling adds the decorrelation step Bagging lacks.
- Top-12 features captures ~80% of cumulative importance.
- CV confirms improvement holds across multiple train/val splits.

Q11 — ANN on USA (target: CChurches)

Train n=259, Test n=111



ANN Setup & Performance

Country selection: USA (370 obs, the most-observed country)

Target chosen: CChurches (most balanced & most predictable from Q5–Q7)

Predictors used: 31 (same as previous classifiers; Country excluded)

Pre-processing for ANN:

- 70 / 30 train / test split within country (seed=12345678).
- StandardScaler applied: each feature z-scored on training data only, then test set transformed with the same scaler. ANNs are sensitive to feature scale; standardisation helps Adam optimiser converge.
- No further encoding needed (all predictors already numeric).

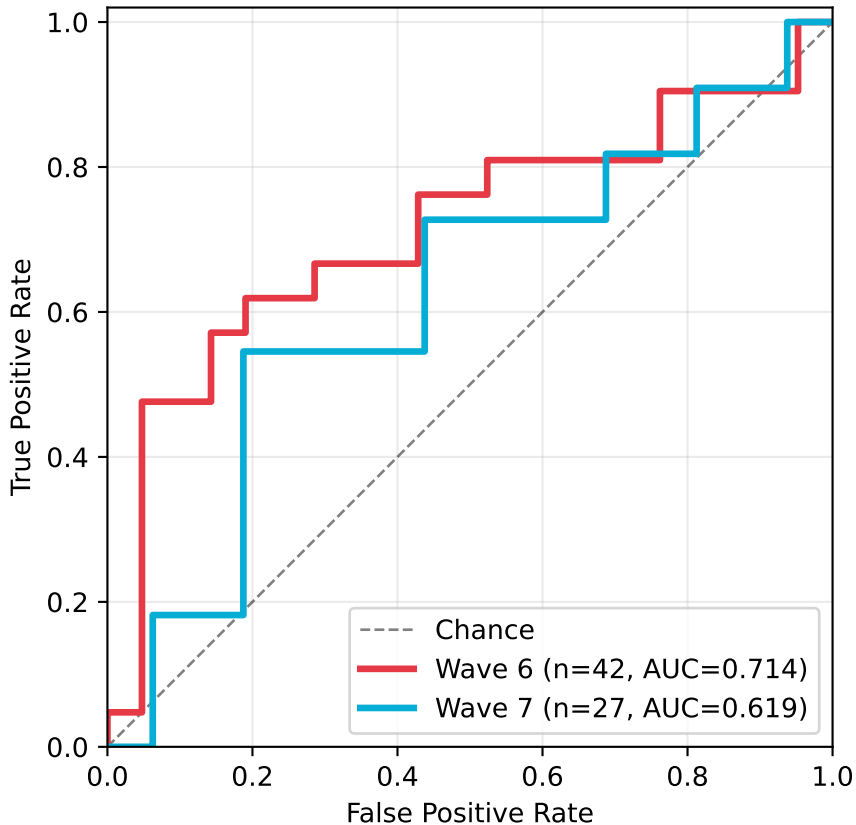
Architecture:

- sklearn MLPClassifier
- Hidden layers: (32, 16) – two layers of ReLU units
- Output: sigmoid (2-class softmax)
- Solver: Adam (adaptive moment estimation)
- max_iter=500, early_stopping=True (validation_fraction=0.15)
- Random seed: 12345678

Performance on test set:

- AUC = 0.6399
- F1-score = 0.6818
- Accuracy = 0.6216

Q12 — ANN Performance by Wave (USA)



Wave	n	Class (0/1)	AUC	F1	Accuracy
Wave 6	42	0:21 / 1:21	0.7143	0.6829	0.6905
Wave 7	27	0:16 / 1:11	0.6193	0.5714	0.5556

Selected waves: top 2 by test obs count in USA.

Interpretation:

- Differences in AUC between waves reflect how respondent attitudes about institutional confidence evolved over time (or sampling variation).
- Lower AUC in one wave suggests the relationship between predictors and CChurches was less stable or more noisy in that period.
- The same trained ANN is applied to both waves so any difference is attributable to data shift, not model change.

Caveats:

- Small-sample noise – wave-level test sets are much smaller than the overall country test set; AUC estimates are higher variance.
- Imbalanced classes within a wave can also shrink AUC differences.
- A formal statistical comparison would require bootstrap CIs on AUC.